

Share of Wallet Intelligence Engine — Syn Bank

Team **The Independent Variable** · Daniel Mataranyika · Herton Mabongue · Standard Bank × Data School Hackathon 2026

5.0%

SYN BANK SHARE

R3.80tn

ADDRESSABLE FLOW

R3.61tn

UNCAPTURED GAP

R8.68bn

FEE OPPORTUNITY / YR

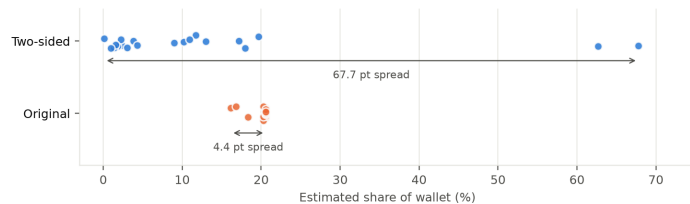
THE PROBLEM, AND THE TRAP IN IT

Syn Bank sees every transaction that flows through Syn Bank — and none of the ones that flow through its competitors. Share of wallet therefore cannot be read off the internal data; it must be estimated from outside.

Our first model missed this. It sized the wallet from internal flow, then took captured volume as a fraction of that wallet:

$$\text{share} = (\text{wallet} \times \text{capture_ratio}) / \text{wallet} = \text{capture_ratio}$$

The wallet term cancels. Share was identical to the assumed capture rate and could not depend on the data. It passed every obvious check — values between 0 and 1, a plausible ranking — while giving **20 very different companies the same answer**: 4.4 points of spread across a portfolio whose wallets differ by 301×.



THE TWO-SIDED MODEL

The two sides of the ratio are now computed from independent sources.

$$\text{share} = \text{captured flow (measured)} / \text{addressable flow (estimated)}$$

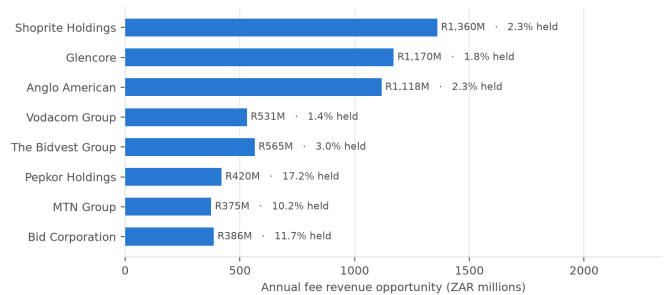
- Top-down** — addressable flow sized from each client's published FY2025 statements: revenue drives transactional, foreign revenue drives FX, cost of sales plus inventory drives trade finance. Each scaled by an explicit SA-attribution share, since Syn Bank is a South African bank.
- Bottom-up** — captured flow summed over the trailing 12 months of the internal datasets. No assumption enters this side.

Share now spans **0.1% to 67.8%** (67.7 points). Every external figure is cited per client in data/external/company_financials.csv.

WHAT A COVERAGE BANKER DOES WITH IT

#	CLIENT	SECTOR	SHARE	FEE OPPTY/YR	LEAD WITH
1	Shoprite Holdings	Consumer Goods	2.3%	R1,360M	Transactional
2	Glencore	Mining	1.8%	R1,170M	Transactional
3	Anglo American	Mining	2.3%	R1,118M	Transactional
4	Vodacom Group	Telecoms	1.4%	R531M	Transactional
5	The Bidvest Group	Industrials & Pharma	3.0%	R565M	Transactional

Ranking blends gap size (50%) with ease of conversion, relationship depth and 90-day urgency, so the list is ordered by winnable value rather than client size. **Transactional banking is the lead pillar for 19 of 20 clients** — cash-management consolidation is the repeatable conversation.



GENERATIVE AI, DOING REAL WORK

The model never supplies a number. Figures are computed by the engine, passed in as an evidence block, and reused verbatim — the model does language, not arithmetic. Every output is then re-read by a grounding check that flags any rand amount or percentage absent from the evidence. It writes pre-meeting briefings and answers open questions over the portfolio, and degrades to deterministic text if the API is unavailable, so a demo never breaks. Prompts and outputs: prompts/logs/.

HOW MUCH DO THE ASSUMPTIONS MATTER?

Shifting every flow multiple by ±25% moves portfolio share only between **4.0% and 6.7%**; the captured side cannot move, because it is measured. Our multiples imply banking fees of ~0.47% of revenue, inside the 0.4–0.7% range the brief itself cites.

LIMITATIONS

20 clients supplied, not the 50 described. Flow multiples are benchmarks, not observations. SA-attribution is judgmental. One revenue figure (Sanlam) is an estimate, flagged and confidence-penalised. Gaps are attributed to competitors, assuming internal records are complete. Lending and DCM are out of scope.